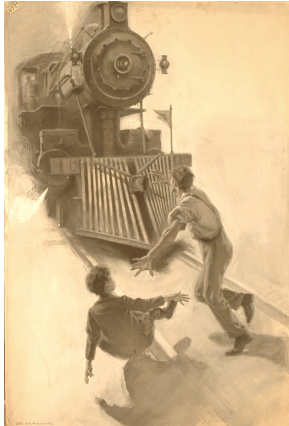


The Simplicity & Beauty of Dynamic Symmetry

– Visual Glossary

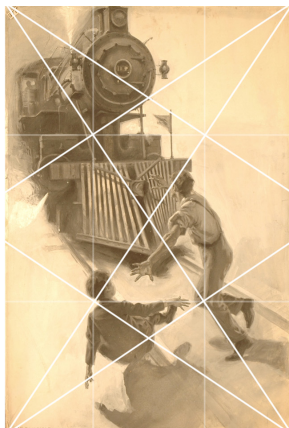
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Artwork by Jay Hambidge

“Instinctive art without mental control is bound to fail” Jay Hambidge.

Before we go any further, I think we should cover where we are getting the term dynamic symmetry and how the armature of our rectangle was built. I would call Jay Hambidge the grandfather of dynamic symmetry. He writes a book that covers it in great detail...though he doesn't show how it relates to art. That's where the canon of design comes into play.



Artwork by Jay Hambidge

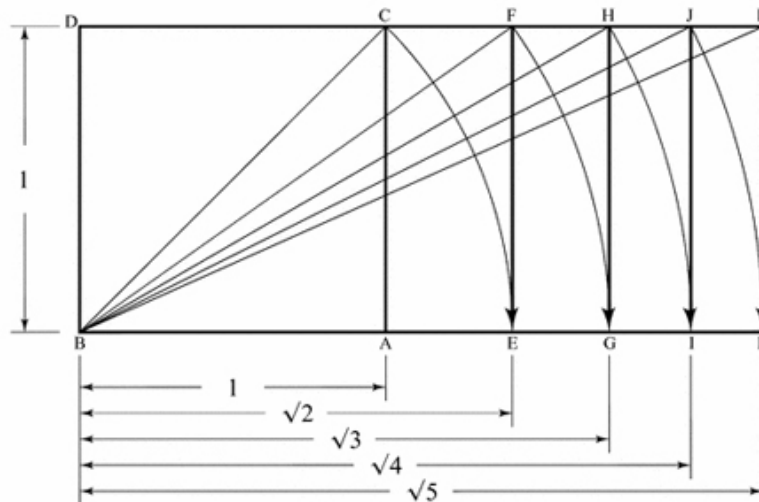
With just the basic armature, we can see how Jay Hambidge is lining up his subjects with the major diagonals, and using his gamut to echo them.

Below I'll introduce you to a handful of rectangles which you can use to design your art within. Knowing how to build each will help you decide which one you will need in the future. because your rectangle's armature and shape should compliment your subjects structure. Have a good idea of what your subject is, then choose a dynamic rectangle to fit it in...not the other way around.

Dynamic Symmetry Root Rectangles

Here is a simple diagram of how to build each root rectangle. It all begins with a square. Swing the diagonal of the square down to make a root 2. Swing the diagonal of the root 2 down to make a root 3....and so on. Easy right?

After a root 5, things start getting rather long, and difficult to apply...but if you need to compose a long panorama shot, you could construct a root 6 or 7 using the same method.



WHAT THE RATIOS ARE USED FOR

Below you'll see ratios next to the root rectangles. Don't let this freak you out! You won't even really need these numbers unless you want to make a canvas or image a specific size. For instance, if you have a 24x36 that you bought at the art store and you want to cut it down to a root 3 rectangle. You like the length of 36 inches, so all you need to do is cut it down according to the root 3 ratio, 1.732.

$$36\text{in} / 1.732 = 20.78 \text{ inches}$$

The same works for centimeters if it's easier for you to measure.

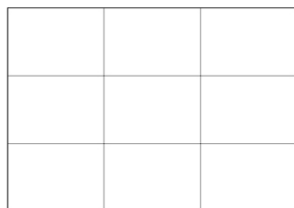
Your final canvas size will be 20.78in x 36in which will equal your root 3 rectangle. What does that mean? Well if you want to shoot for a specific frame size, you will know which rectangle to use with the minimal amount of cropping.

Root 2 Rectangle= 1.414, very close to 5x7

We can easily break down the Root 2 rectangle on the theme of two or three. Theme, meaning divide it in half or in thirds. But remember, we are still going to build the basic armature within each rectangle.



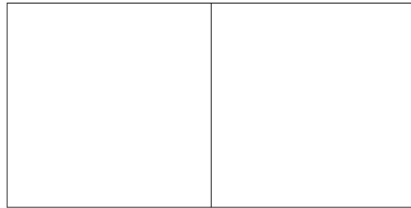
ROOT 2 RECTANGLE
(theme of 2)



ROOT 2 RECTANGLE
(theme of 3)

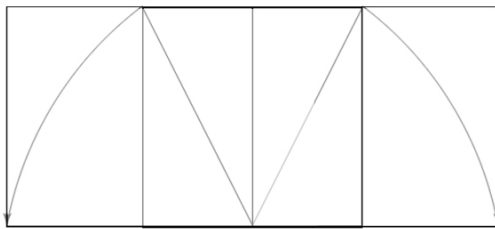
Root 3 Rectangle = 1.732 very close to the 16:9 cinema screen

Root 4 Rectangle = 2 *(the only rectangle with a rational number)*



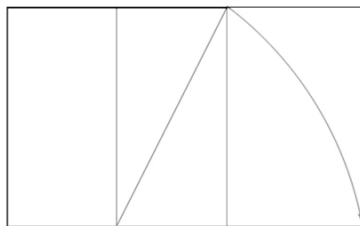
ROOT 4 RECTANGLE
(two squares side by side)

Root 5 Rectangle = 2.236 very close to the 21:9 ultra wide cinema screen

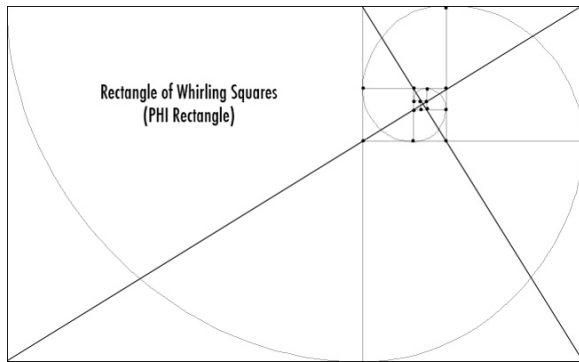


ROOT 5 RECTANGLE
(two overlapping PHI's making a square in the middle)

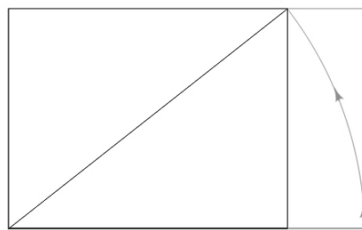
Phi Rectangle- 1.618 This is the rectangle of the whirling squares. The ratio that is found in the human body and nature. This is close to the 1.5 ratio rectangle, so if you had to, you could use the frame sizes for that and not crop anything. *(I always like to make it the size of the rectangle I chose to shoot because the dynamic symmetry is specific to the rectangle. Cropping your image differently than the rectangle you composed with will violate the integrity of the dynamic symmetry...but if you are trying to match a certain frame you will have to something custom to fit, or make a compromise.)*



PHI RECTANGLE 1.618

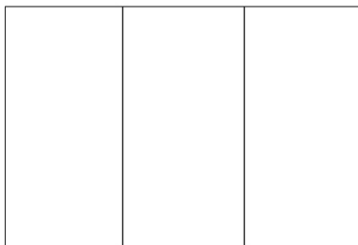


Root Phi Rectangle- 1.2720 Take the length of the PHI rectangle, swing it up from the bottom and this will give you the Root PHI. It' is the smallest of all of them, and it is almost exactly the same ratio as the standard 11x14 frame size, and very close to 8.5x11, 14x18 and 28x22.



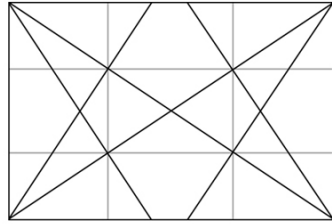
ROOT PHI RECTANGLE 1.2720

1.5 Rectangle - a square and a half. Equal to 4x6, 8x12 or 24x36 frame size



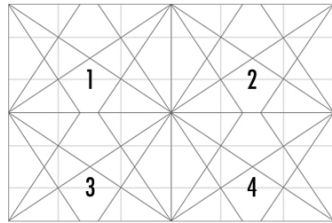
1.5 RECTANGLE (square and a half)

As we covered in the Canon of Design Essentials, the basic armature of a rectangle consists of two diagonals, and four reciprocals. Then you can add horizontals and verticals through the eyes of each intersection. That's it. Easy!



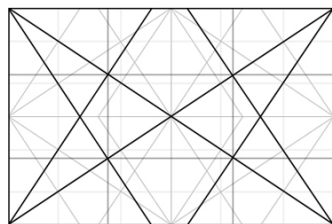
1.5 RECTANGLE with Basic Armature
(with added Verticals and Horizontals)

When you are building your composition, say a painting, or a well laid out photograph with set design, you can break it down even further using major area divisions (MAD). Just use the same basic armature of the original rectangle, and shrink it down to make 4 more.



1.5 RECTANGLE with Basic Armature
(with four smaller MAD)

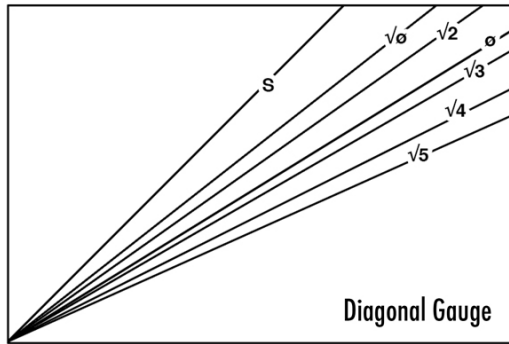
Here is the basic armature (dark) with the underlying MAD (lighter). This is the technique you can use for composing your work as well as analyzing other works of art.



1.5 RECTANGLE with MAD

You can use this basic armature build up with all of the rectangles (even arbitrary rectangles if you really must).

The gauge below can be used on unknown rectangles to find the diagonal easily, then you'll know exactly which one was being used.



Cropping your image differently than the rectangle you composed with will violate the integrity of the dynamic symmetry.

The Rule of thirds won't give us variety, unity, and order like dynamic symmetry and design will.

"Design created within rectangles which do not possess dynamic symmetry, the qualities of life and growth, are always flat and dead." Jay Hambidge

